

RESEARCH PAPER

Customer Segmentation & Churn Pattern Analytics in European Banking

A Segmentation-Driven and Predictive Analytics Study of Customer Retention

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Dataset & Scope

This study analyses 10,000 retail banking customer records across three European markets — France, Germany, and Spain — covering demographic, financial, and engagement attributes, with the binary outcome of customer attrition (churn) as the target variable. All figures reported in this paper were computed directly from the underlying dataset and cross-verified against the accompanying Streamlit analytics dashboard.

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Abstract

Customer churn represents one of the most significant sources of preventable revenue loss in retail banking. This study analyses a portfolio of 10,000 customers across France, Germany, and Spain to identify the demographic, financial, and behavioural factors most strongly associated with attrition. Using a segmentation-driven analytical methodology, the portfolio is examined across seven dimensions — geography, age, credit standing, tenure, account balance, product holding, and engagement status — revealing an overall churn rate of 20.37% against a retention rate of 79.63%.

The analysis identifies material risk concentrations that are obscured when any single dimension is examined in isolation. Most notably, the interaction between geography and age reveals that customers aged 46–60 in Germany churn at 67.33%, more than three times the portfolio average, despite neither Germany nor the 46–60 age band appearing exceptional when analysed independently. Engagement status is confirmed as a leading indicator, with inactive customers churning at 26.85% versus 14.27% for active customers, a gap of 12.58 percentage points. Product holding emerges as the strongest single behavioural signal, with customers holding four products churning at 100% and those holding one product churning at 27.71%, against a low of 7.58% for two-product holders.

To extend this analysis beyond descriptive segmentation, a Gradient Boosting classifier was trained and validated using five-fold cross-validation, achieving a Receiver Operating Characteristic Area Under the Curve (ROC-AUC) of 0.866 — materially outperforming a Logistic Regression baseline (ROC-AUC 0.755). Applied to the full portfolio, the model identifies 991 customers (9.9%) as High Risk, a group that churns at an actual rate of 89.30%, representing a risk concentration that segmentation alone cannot achieve. The paper concludes with a set of prioritised, segment-specific retention recommendations and a proposed marketing action framework based on a customer value × engagement matrix.

1. Introduction and Background

Customer churn is among the largest hidden costs in retail banking. Each departing customer represents not only the direct loss of that customer's lifetime value, but also an increase in the acquisition cost required to replace them, and a source of instability in projected revenue. While most banks track an aggregate churn rate as a standing operational metric, this figure alone provides limited guidance for action: it does not indicate which customer groups are most exposed, how attrition patterns differ across geography, age, or financial standing, or whether churn is concentrated among high-value or low-value relationships.

This paper addresses that gap through a systematic, segmentation-driven analysis of a 10,000-customer retail banking portfolio spanning three European markets. The analysis proceeds from descriptive segmentation through to predictive modelling, with the explicit goal of producing findings that translate directly into retention strategy rather than remaining purely diagnostic.

2. Problem Statement

Despite holding rich, customer-level data, retail banks frequently lack the structured analytical framework needed to convert that data into retention strategy. Three specific gaps motivate this study:

1. Identification of high-risk customer segments across multiple, simultaneously-considered dimensions rather than single-variable summaries.
2. Understanding of how churn differs by geography, age, and financial profile, including interaction effects that are invisible when dimensions are analysed separately.
3. Quantification of the financial profile of churned customers, distinguishing high-value attrition (which carries outsized revenue impact) from ordinary attrition.

Without a structured analytical response to these gaps, retention strategy tends toward generic, reactive campaigns rather than targeted, proactive intervention.

3. Objectives

3.1 Primary Objectives

- Measure the overall churn rate for the full customer portfolio.
- Identify how churn is distributed across customer segments defined by geography, age, credit standing, tenure, and account balance.
- Compare churn behaviour across the three European regions represented in the dataset.

3.2 Secondary Objectives

- Understand the concentration of churn among high-value customers and quantify the associated revenue exposure.
- Evaluate engagement and tenure patterns across the customer lifecycle, including single-year effects obscured by grouped tenure bands.
- Support strategic planning and marketing decision-making through a segment-specific action framework.
- Extend descriptive segmentation with a predictive model capable of scoring individual customers by churn risk, addressing a limitation identified in prior analytical work on this customer base.

4. Dataset Description

The dataset comprises 10,000 customer records with thirteen analytically relevant fields, summarised below. Two fields present in the raw export — Surname and a constant Year field — were removed prior to analysis: Surname carries no analytical relationship to churn, and Year was found to hold a single value across all 10,000 rows, contributing no discriminative information.

Field	Description
CustomerId	Unique customer identifier
CreditScore	Customer creditworthiness (numeric)
Geography	Country of residence: France, Germany, or Spain
Gender	Male or Female
Age	Customer age in years
Tenure	Years as a customer of the bank (0–10)
Balance	Account balance
NumOfProducts	Number of bank products held (1–4)
HasCrCard	Credit card ownership (binary)
IsActiveMember	Activity indicator (binary)
EstimatedSalary	Estimated annual salary
Exited	Churn indicator — the target variable (binary)

Table 4.1 — Dataset field description, post-cleaning.

The dataset exhibits no missing values across any field, confirming data quality sufficient for direct analysis without imputation. The target variable is moderately imbalanced: 2,037 customers (20.37%) exited, against 7,963 (79.63%) retained — a ratio addressed explicitly in the predictive modelling stage (Section 7).

5. Methodology

5.1 Data Cleaning and Preparation

Following removal of the Surname and Year fields, five derived segmentation fields were engineered from the raw variables to support the multi-dimensional analysis in Section 6:

Derived Field	Basis
Age Group	<30, 30–45, 46–60, 60+ — matching the bands specified in the project brief
Credit Band	Low (<580), Medium (580–700), High (700+) — standard FICO-style credit risk thresholds
Tenure Group	New (0–2 yr), Mid-term (3–6 yr), Long-term (7–10 yr)
Balance Segment	Zero-balance, Low-balance (€0–100,000), High-balance (€100,000+)
High-Value Customer	Top quartile by Balance OR top quartile by Estimated Salary, computed on the full portfolio

Table 5.1 — Derived segmentation fields and their construction logic.

Credit score thresholds follow conventional credit-risk banding rather than arbitrary quantile splits, reflecting industry practice. The zero-balance segment is deliberately separated from the low-balance segment, since a customer holding literally no funds represents a materially different risk profile than one with a modest positive balance.

5.2 Segmentation Design

Seven segmentation dimensions were defined for the analysis: Geography, Gender, Age Group, Credit Band, Tenure Group, Balance Segment, and Activity Status. Each dimension supports two complementary metrics — the within-segment churn rate (the probability that a customer in that segment churns) and the segment's contribution to total churn volume (the share of all churned customers originating from that segment). This distinction is treated as analytically important throughout the paper: a segment can carry a high churn rate while contributing little to total churn volume if the segment is small, and conversely a large segment with a moderate churn rate can still be the dominant source of total attrition.

5.3 Key Performance Indicator Framework

KPI	Definition
Overall Churn Rate	Percentage of customers who exited
Segment Churn Rate	Churn percentage within a defined segment
High-Value Churn Ratio	Churn rate among top-quartile-value customers
Geographic Risk Index	Regional churn rate divided by the portfolio-wide churn rate
Engagement Drop Indicator	Percentage-point gap between inactive and active customer churn rates

Table 5.2 — Key Performance Indicators used throughout this study.

5.4 Predictive Modelling Approach

Four candidate classifiers — Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting — were benchmarked on an 80/20 stratified train-test split, with results confirmed via five-fold stratified cross-validation to guard against a favourable single split. Ten input features were used: Credit Score, Geography, Gender, Age, Tenure, Balance, Number of Products, Credit Card Ownership, Activity Status, and Estimated Salary. Model selection is detailed in Section 7.

6. Exploratory Data Analysis and Findings

6.1 Overall Churn Summary

Across the full portfolio of 10,000 customers, 2,037 exited and 7,963 were retained, yielding an overall churn rate of 20.37% and a retention rate of 79.63%. This baseline figure anchors every segment-level comparison in the remainder of this section.

6.2 Geographic Analysis

Geography	Customers	Churned	Churn Rate	Risk Index
France	5,014	810	16.15%	0.79×
Germany	2,509	814	32.44%	1.59×
Spain	2,477	413	16.67%	0.82×

Table 6.1 — Churn rate and Geographic Risk Index by region.

Germany presents materially elevated risk relative to France and Spain, churning at 32.44% against a portfolio average of 20.37% — a Geographic Risk Index of 1.59×. France and Spain, by contrast, both churn below the portfolio average, at 0.79× and 0.82× respectively. Despite holding fewer customers than France (2,509 versus 5,014), Germany contributes almost exactly as many churned customers in absolute terms (814 versus 810), underscoring that Germany's elevated rate is not offset by its smaller base.

6.3 Demographic Analysis: Age and Gender

Age Group	Customers	Churn Rate
<30	1,968	7.52%
30–45	5,921	15.74%
46–60	1,647	51.12%
60+	464	24.78%

Table 6.2 — Churn rate by age group.

Churn risk rises sharply with age through the 46–60 band, which churns at 51.12% — more than 2.5 times the portfolio average — before declining somewhat in the 60+ band to 24.78%. This non-monotonic pattern suggests the 46–60 band represents a distinct risk period, plausibly associated with pre-retirement financial reassessment, rather than churn risk simply increasing with age throughout the customer lifecycle.

Gender-based analysis shows Female customers churning at 25.07% against 16.46% for Male customers, a gap of 8.61 percentage points across 4,543 and 5,457 customers respectively.

6.4 Geography × Age Interaction Effect

Rank	Geography	Age Group	Churn Rate
1	Germany	46–60	67.33%
2	France	46–60	45.79%
3	Spain	46–60	40.66%

Table 6.3 — Highest-churn Geography × Age Group combinations.

This is the single most consequential finding in the descriptive portion of this study. German customers aged 46–60 churn at 67.33%, more than three times the portfolio-wide baseline of 20.37%, and substantially exceeding what either dimension predicts independently: Germany overall churns at 32.44%, and the 46–60 age band overall churns at 51.12%. Neither figure alone anticipates the 67.33% observed when the two dimensions intersect. This interaction effect is structurally invisible to any analysis that examines geography and age separately, and demonstrates the necessity of multi-dimensional segmentation over single-variable summary statistics.

6.5 Tenure and Engagement Patterns

Tenure Group	Customers	Churn Rate
New (0–2 yr)	2,496	21.15%
Mid-term (3–6 yr)	3,977	20.64%
Long-term (7–10 yr)	3,527	19.51%

Table 6.4 — Churn rate by tenure group.

Grouped into three bands, tenure shows only a mild downward trend in churn risk as the relationship lengthens. A year-by-year examination, however, reveals a sharper effect that grouped bands conceal: Tenure Year 0 registers the single highest churn rate of any individual year, at 23.00%, followed by Year 1 at 22.42%. This early-relationship elevation — commonly termed ‘honeymoon churn’ in retail banking — points to onboarding and first-year experience as a distinct intervention point, separate from the more gradual, multi-year loyalty dynamics that grouped tenure bands are better suited to describe.

Engagement status is confirmed as a materially stronger predictor than tenure alone. Active customers (5,151 of the portfolio) churn at 14.27%, while inactive customers (4,849) churn at 26.85% — an Engagement Drop Indicator of 12.58 percentage points, positioning activity status as one of the most actionable signals identified in this study, since it is directly observable and, unlike age or geography, addressable through re-engagement campaigns.

6.6 Financial Stability Indicators

Indicator	Segment	Churn Rate
Credit Score Band	Low (<580)	22.15%
Credit Score Band	Medium (580–700)	19.77%
Credit Score Band	High (700+)	19.87%
Number of Products	1 product	27.71%
Number of Products	2 products	7.58%
Number of Products	3 products	82.71%
Number of Products	4 products	100.00%
Credit Card Ownership	No card	20.81%
Credit Card Ownership	Has card	20.18%

Table 6.5 — Churn rate by financial stability indicator.

Credit score band shows only a weak relationship with churn, and credit card ownership shows almost none — both indicators cluster tightly around the portfolio average, suggesting neither is a meaningful standalone predictor. Product holding, by contrast, is the most striking financial indicator in the dataset: churn rises from 27.71% among

single-product customers, drops sharply to a portfolio-low 7.58% among two-product customers, then rises steeply to 82.71% among three-product holders and 100% among the small four-product cohort (60 customers).

This U-shaped, then sharply rising, pattern is interpreted as two distinct mechanisms rather than one. The elevated churn among single-product customers reflects low switching costs and shallow relationship depth — these customers have little invested in the relationship and correspondingly little friction to leaving. The near-universal churn among three- and four-product customers is more consistent with an over-selling dynamic, in which products were added faster than genuine need, producing customers who feel mis-sold rather than loyal. The two-product segment, which minimises churn, plausibly represents the point of genuine, sustainable product-market fit for the majority of the customer base.

6.7 Comparison of Churned versus Retained Profiles

Metric	Retained (avg)	Churned (avg)	Difference
Credit Score	651.9	645.4	−1.0%
Age	37.4	44.8	+19.9%
Tenure	5.0	4.9	−2.0%
Balance	€72,745	€91,109	+25.2%
Number of Products	1.5	1.5	−4.5%
Estimated Salary	€99,738	€101,466	+1.7%

Table 6.6 — Average profile: retained versus churned customers.

Balance shows the widest proportional gap between the two groups (+25.2%), followed by Age (+19.9%). Churned customers hold materially higher account balances on average, which — combined with the High-Value Customer findings in Section 6.8 — reinforces that this portfolio's attrition is not concentrated among low-value, disengaged customers alone, but meaningfully involves financially substantial relationships as well.

6.8 High-Value Customer Analysis

High-value customers, defined as the top quartile by account balance or estimated salary computed across the full portfolio, number 4,351 — 43.5% of the customer base. This group churns at 22.27%, moderately above the 18.91% rate observed among standard customers. While the rate differential is moderate, the revenue consequence is not: churned high-value customers collectively represent €113,353,534 in lost account balance, a figure that substantially understates the true relationship value once cross-sell, lifetime fee revenue, and referral effects are considered.

Section 6 Summary

The descriptive segmentation stage identifies engagement status, geography (particularly Germany), the 46–60 age band, and product holding as the four strongest individual churn signals, with the Germany × 46–60 interaction and the early-tenure spike as the two most actionable multi-dimensional findings. Section 7 extends this analysis with a predictive model capable of combining all signals simultaneously.

7. Predictive Modelling

7.1 Model Selection

Four classifiers were benchmarked on an identical 80/20 stratified train-test split of the ten-feature input set, with results confirmed via five-fold stratified cross-validation:

Model	Accuracy	Precision	Recall	F1	ROC-AUC (CV)
Logistic Regression	80.5%	0.59	0.14	0.23	0.755
Decision Tree	85.4%	0.79	0.39	0.52	—
Random Forest	82.2%	0.55	0.72	0.62	0.859
Gradient Boosting	87.0%	0.79	0.49	0.61	0.866

Table 7.1 — Candidate model comparison.

Logistic Regression is disqualified as a primary model despite an acceptable ROC-AUC of 0.755: its recall of 0.14 means it correctly identifies only 14% of customers who actually churn, a failure mode that directly undermines the purpose of a churn model. Gradient Boosting was selected as the primary model on the strength of its leading ROC-AUC (0.866) and its balance of precision and recall, with Random Forest retained for its superior recall (0.72) where the operational priority is minimising missed at-risk customers rather than minimising false positives.

Gradient Boosting's advantage over the linear baseline is consistent with the segmentation findings in Section 6: the Germany × Age interaction effect (Section 6.4) is, by construction, a non-linear relationship between two variables, which a linear model structurally cannot represent as effectively as a tree-based ensemble.

7.2 Feature Importance

Feature	Relative Importance
Age	36.5%
Number of Products	26.0%
Balance	9.9%
Is Active Member	6.5%
Estimated Salary	5.5%
Credit Score	5.3%
Geography	4.4%
Tenure	2.8%
Gender	2.2%
Has Credit Card	0.7%

Table 7.2 — Random Forest feature importance.

Age and Number of Products jointly account for 62.5% of the model's predictive weight, independently confirming the two strongest descriptive findings from Section 6 (the age-driven interaction effect and the product-holding U-curve) through an entirely separate analytical method. This convergence between manual segmentation and model-derived feature importance strengthens confidence in both approaches.

7.3 Risk Segmentation Using the Predictive Model

Applying the trained model to the full, unfiltered portfolio and grouping customers into Low, Medium, and High risk bands (0–30%, 30–60%, 60–100% predicted probability) yields the following distribution:

Risk Band	Customers	Actual Churn Rate
Low Risk	7,885	7.89%
Medium Risk	1,124	47.15%
High Risk	991	89.30%

Table 7.3 — Model-based risk segmentation, full portfolio.

The 991 customers (9.9% of the portfolio) flagged as High Risk churn at an actual rate of 89.30%, an 11.3× concentration relative to the portfolio-wide baseline of 20.37%. This concentration substantially exceeds what any single-dimension or two-dimension manual segment achieves elsewhere in this study — including the Germany × 46–60 interaction (67.33%) — demonstrating the practical value of combining all ten input variables simultaneously rather than relying on manually-selected dimension pairs.

8. Key Findings Summary

1. Overall portfolio churn stands at 20.37%, with retention at 79.63%, across 10,000 customers.
2. Germany is the highest-risk market, churning at 32.44% — a Geographic Risk Index of 1.59× — versus 0.79× for France and 0.82× for Spain.
3. The 46–60 age band churns at 51.12%, more than 2.5× the portfolio average, with risk declining in the 60+ band.
4. The Germany × 46–60 interaction produces a 67.33% churn rate — the single highest risk concentration identified through descriptive segmentation — a result neither dimension predicts in isolation.
5. Engagement status is a leading indicator: inactive customers churn at 26.85% versus 14.27% for active customers, a 12.58-point gap.
6. Tenure Year 0 shows the highest single-year churn rate (23.00%), indicating an early-relationship ‘honeymoon churn’ effect obscured by grouped tenure bands.
7. Product holding is the strongest financial stability signal: churn ranges from 7.58% (2 products, the optimum) to 100% (4 products), with 1-product and 3–4-product customers both elevated for different underlying reasons.
8. High-value customers (43.5% of the portfolio) churn at 22.27%, exposing €113,353,534 in lost account balance among those already churned.
9. Balance shows the widest proportional difference between churned and retained customer profiles (+25.2%), followed by Age (+19.9%).
10. A Gradient Boosting model (ROC-AUC 0.866) outperforms Logistic Regression (ROC-AUC 0.755) and identifies a High Risk group of 991 customers churning at 89.30% — an 11.3× concentration exceeding any manual segment identified in this study.

9. Strategic Recommendations

9.1 Customer Value × Engagement Action Matrix

Combining customer value tier (High-Value / Standard) with engagement status (Active / Inactive) yields four segments, each warranting a distinct marketing response:

Segment	Recommended Action
High-Value × Inactive	Priority win-back — dedicated relationship-manager outreach, given the combination of elevated revenue-per-customer and elevated risk.
High-Value × Active	Loyalty rewards and cross-sell campaign to deepen an already-engaged, high-value relationship.
Standard × Inactive	Automated re-engagement campaign, prioritised by predicted risk score for efficient resource allocation.
Standard × Active	Standard nurture marketing; lowest priority for incremental retention spend.

Table 9.1 — Recommended marketing action by customer value × engagement segment.

9.2 Segment-Specific Recommendations

1. Direct retention campaign design specifically at German customers aged 46–60, rather than applying a uniform regional strategy — this single interaction segment carries disproportionate risk concentration relative to its size.
2. Redesign the first-year onboarding programme with the Year 0 attrition spike specifically in mind; consider a structured 90-day and 12-month engagement checkpoint distinct from the standing retention programme aimed at longer-tenured customers.
3. Route customers transitioning from Active to Inactive status into an automated retention workflow immediately upon status change, rather than waiting for a churn event to trigger outreach.
4. Investigate the three- and four-product customer segment for a potential over-selling pattern by cross-referencing product addition dates against sales channel and bundling practices; the near-universal churn in this segment warrants a service-quality review rather than further cross-sell.
5. Establish a dedicated relationship-manager coverage track for high-value customers, separate from the general retention programme, given the €113.4 million revenue exposure already realised among churned high-value customers.
6. Adopt the Gradient Boosting model's ChurnRiskScore as the primary prioritisation mechanism for retention outreach, reserving the 991-customer High Risk list for immediate, resource-intensive intervention and the Medium Risk group (1,124 customers) for the standard retention cadence.

10. Conclusion

This study demonstrates that effective churn management in retail banking requires moving beyond single-variable summary statistics toward genuinely multi-dimensional segmentation. The portfolio's overall churn rate of 20.37% conceals substantial internal variation: risk ranges from under 8% among engaged, two-product, Low-Risk-scored customers to over 89% among the model's High Risk segment, and single interaction effects — most notably Germany × 46–60 — produce risk concentrations invisible to any analysis confined to one dimension at a time.

The extension of this analysis from descriptive segmentation to predictive modelling is shown to add material value rather than merely academic interest: the Gradient Boosting model's 991-customer High Risk segment achieves an 11.3× risk concentration, exceeding every manually-constructed segment examined in this study, while its feature importance ranking independently corroborates the two strongest findings reached through manual segmentation. Together, the descriptive and predictive analyses converge on a consistent picture — age, product holding, engagement status, and geography, particularly in combination, are the dominant drivers of attrition in this portfolio — and support the segment-specific, prioritised retention strategy set out in Section 9.

10.1 Limitations and Future Work

- The dataset provides a single cross-sectional snapshot; a longitudinal extension tracking individual customers over time would allow the early-tenure and mid-life effects identified in Section 6.5 to be validated against actual behavioural trajectories rather than inferred from cohort comparisons.
- Model recall (49.1%) leaves room for improvement; class-balancing techniques (e.g. SMOTE) or threshold tuning specific to the retention team's operational capacity were outside the scope of this study and are recommended as a next iteration.
- The High-Value Customer definition relies on a fixed top-quartile threshold; a continuous customer lifetime value model would provide finer-grained prioritisation within the current binary High-Value / Standard split.
- Product-holding causality (Section 6.6) is inferred from cross-sectional churn rates; a controlled analysis of product addition sequencing and timing relative to churn events would substantiate the proposed over-selling and low-embeddedness mechanisms more directly.

Appendix A: Key Performance Indicator Reference

KPI Name	Description	Portfolio Value
Overall Churn Rate	% customers who exited	20.37%
Segment Churn Rate (highest)	Churn % by segment	51.12% (Age 46–60)
High-Value Churn Ratio	Churn among premium customers	22.27%
Geographic Risk Index (highest)	Regional churn exposure	1.59× (Germany)
Engagement Drop Indicator	Inactivity vs. active churn gap	12.58 points

Table A.1 — Full KPI reference, computed on the complete, unfiltered portfolio.

Note on Reproducibility

All figures in this paper were computed directly from the underlying 10,000-row dataset using the same segmentation logic, KPI definitions, and model configuration (Gradient Boosting, `n_estimators=200`, `max_depth=3`, `random_state=42`; train/test split 80/20, stratified, `random_state=42`) implemented in the accompanying Streamlit dashboard, and were independently cross-verified against the dashboard's live outputs prior to submission.